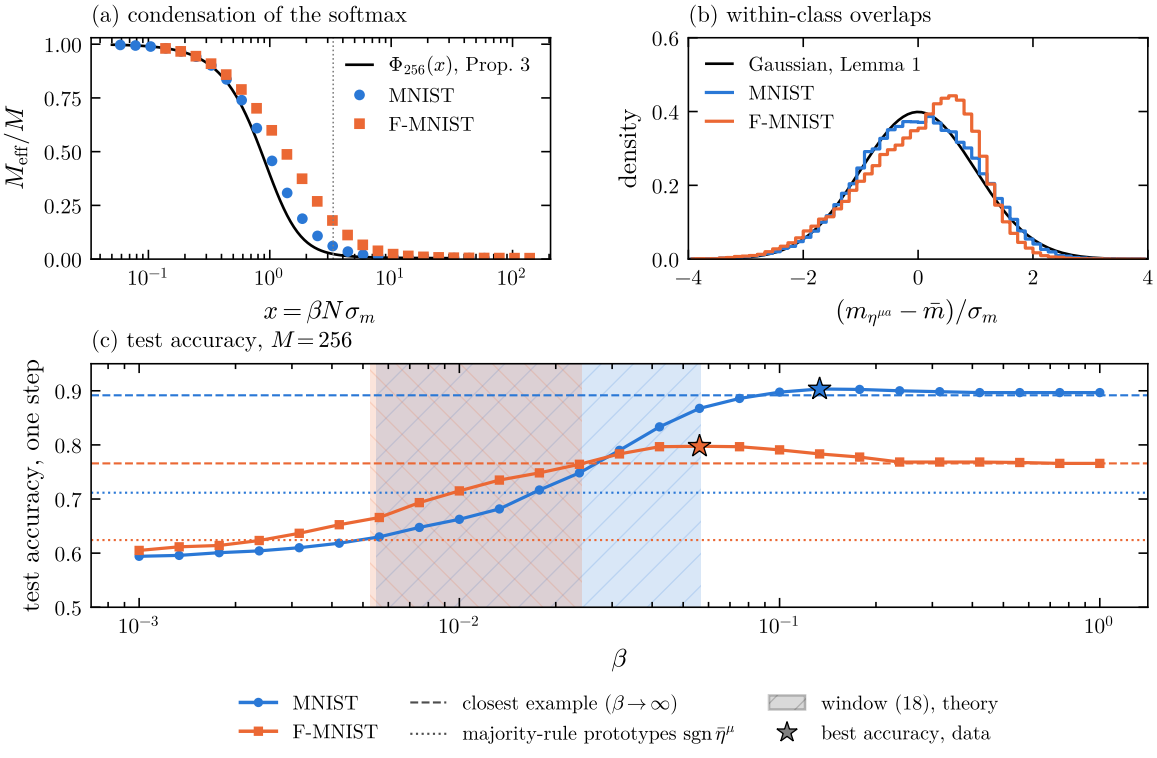
(a) condensation of the softmax
M_{eff}/M
1.00
0.75
0.50
0.25
0.00
10^{-1}
10^{0}
10^{1}
10^{2}
$x = \beta N \sigma_m$
$\Phi_{256}(x)$, Prop. 3
MNIST
F-MNIST
(b) within-class overlaps
density
0.6
0.4
0.2
0.0
−4
−2
0
2
4
$(m_{\eta^{\mu a}} - \bar{m})/\sigma_m$
Gaussian, Lemma 1
MNIST
F-MNIST
(c) test accuracy, $M = 256$
test accuracy, one step
0.9
0.8
0.7
0.6
0.5
10^{-3}
10^{-2}
10^{-1}
10^{0}
β
MNIST
F-MNIST
closest example ($\beta \to \infty$)
majority-rule prototypes $\operatorname{sgn} \bar{\eta}^{\mu}$
window (18), theory
best accuracy, data